

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, LUCKNOW

A Data-Driven Framework for Wind Energy Deviation Penalty Optimization

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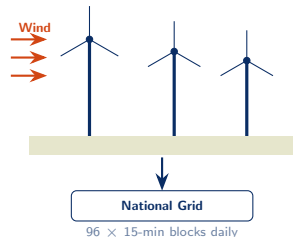
1. Introduction

India's Renewable Energy Context

- ▶ India targets **500 GW** of clean energy by 2030
- ▶ Wind power is critical but **unpredictable** — depends on weather
- ▶ Poor forecast → wrong schedule → **financial penalties**

The Grid Scheduling Requirement

- ▶ All farms must declare output in **96 slots of 15 min each** daily
- ▶ Schedule submitted **1-day** before each slot
- ▶ Deviation from the schedule triggers **CERC DSM charges**



2. Problem Statement & DSM

What is DSM?

The **Deviation Settlement Mechanism (DSM)** is CERC's penalty framework. If actual wind generation deviates by more than **10%** from the declared schedule, escalating charges are applied.

Why Is This Hard?

- ▶ Global weather models are designed for large regions, not single farms
- ▶ Existing average error = **5.96 MW** — near the 10% limit
- ▶ Evening errors (6–10 PM) cost **2.5× more**
- ▶ Result: Crores of rupees in avoidable annual penalties

DSM Penalty Bands (CERC 2024)

0–10%	No penalty — Safe Zone
10–20%	10% of market price
20–30%	20% of market price
>30%	30% of market price

For 75

MW plant: 10% = only **7.5 MW** tolerance

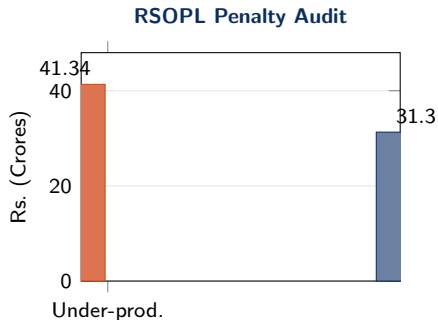
3. Case Study — RSOPL Koppal, Karnataka

Site Details

- ▶ **Plant:** RSOPL, Koppal, Karnataka
- ▶ **Coordinates:** 15.34°N, 76.15°E
- ▶ **Capacity:** 75 MW registered
- ▶ **Turbines:** Vestas class, 3 MW, 100 m hub

10-Month Financial Audit

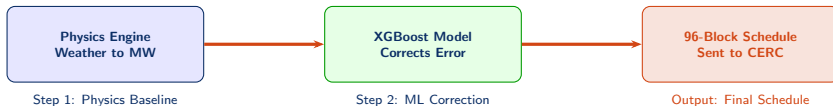
- ▶ Under-production penalties: **Rs. 41.34 Crores**
- ▶ Over-production credits: Rs. 31.30 Crores
- ▶ Avg. schedule error: **5.96 MW (7.95%)**
- ▶ PI-PML targets cutting this penalty significantly



Rs. 41 Crores is our savings target

4. My Solution — PI-PML Framework

Physics-Informed Predictive Machine Learning (PI-PML)



Input
NOAA GFS weather
(Wind, Temp, Pressure)

Trained on
10 months real SRPC
generation records

Goal
Reduce DSM penalties
save Crores annually

5. Solution Definition

Physics Engine

- ▶ **Tetens equation** — humid monsoon air is lighter, less power output
- ▶ **Hellman law** — wind at 100 m is 32% faster than at 10 m ground level
- ▶ **Betz limit** — turbine extracts at most 59.3% of wind energy
- ▶ Rule-based, no training data needed

XGBoost Residual Corrector

- ▶ Physics still misses: turbine wear, curtailment, local turbulence
- ▶ **Residual** = Actual generation – Physics estimate
- ▶ XGBoost is trained to predict this gap
- ▶ **Final:** Physics + XGBoost = corrected schedule
- ▶ Output clipped to range [0, 75 MW]

Forecast Pipeline



6. Dataset & Architecture

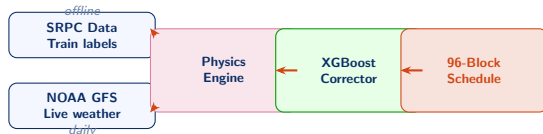
Training Data — SRPC

- ▶ 45 weekly ZIP files from SRPC government portal
- ▶ Python script auto-extracts all files
- ▶ **29,568 blocks**, Apr 2025 – Feb 2026
- ▶ Columns: scheduled MW, actual MW, datetime

Live Weather — NOAA GFS

- ▶ Free API, updated **4 times per day**
- ▶ Wind at native **100 m height** (matches turbine hub)
- ▶ Fetched automatically every morning

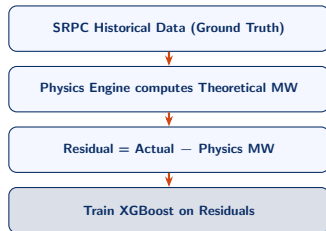
System Architecture



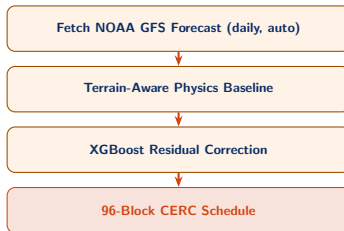
Metric	Value
Total blocks	29,568
Error (before)	5.96 MW
Error (after)	1.71 MW
Improvement	71.2%

7. How It Works

Phase A — Training



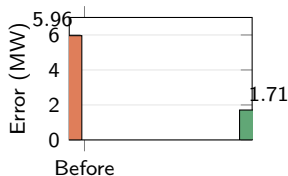
Phase B — Upcoming: Live Day-Ahead Pipeline



8. Conclusion & Upcoming Deliverables

Work Completed

- ▶ CERC DSM 2024 regulatory audit done
- ▶ **Dataset 1:** SRPC — 29,568 blocks collected
- ▶ **Dataset 2:** NOAA GFS identified & tested
- ▶ Physics Engine built (Tetens, Hellman, Betz)
- ▶ XGBoost residual model trained & validated
 - ★ **71.2% error reduction**
 - ★ **Rs. 23.5 Lakhs saved (2-month test)**



Upcoming Deliverables

- ▶ **Automated Weather Ingestion:** Daily NOAA GFS fetch for Koppal coordinates
- ▶ **Terrain-Aware Wind Correction:** Roughness factor for Deccan Plateau hill terrain
- ▶ **Full 96-Block Live Pipeline:** Physics → XGBoost → CERC schedule, automated daily
- ▶ **CERC-Formatted Submission:** DSM-compliant 96-block day-ahead output
- ▶ **Financial Validation Report:** Month-by-month penalty savings vs. = original schedule

Thank You

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